

SONA COLLEGE OF TECHNOLOGY

(AUTONOMOUS INSTITUTION)

DEPARTMENT OF INFORMATION TECHNOLOGY

MINI PROJECT – U23IT705 (2026-27)

DATE:

BATCH NO:	15	
REGISTRATION NUMBER	STUDENT NAME	SIGNATURE
61782323106001	AATHIKESAVAA S	
61782323106016	BHARATH R	
61782323106040	HARSHINI S	
TITLE OF THE PROJECT:	MULTI-PILLAR DEEPPFAKE DETECTION BASED ON VISION TRANSFORMER AND ERROR LEVEL ANALYSIS	
DOMAIN NAME/AREA:	CYBERSECURITY & DIGITAL FORENSICS	
Algorithm/Technique used:	Vision Transformers (ViT), Mel-Spectrogram Extraction, Benford's Law Statistical Analysis,	
REFERENCE (JOURNAL/ ONLINE RESOURCES/ OTHERS)	https://arxiv.org/abs/2010.11929 , https://arxiv.org/abs/2111.02599	
GUIDE NAME & SIGNATURE	MR. A. NAVEEN KUMAR, AP / IT	

SONA COLLEGE OF TECHNOLOGY
(AUTONOMOUS INSTITUTION)
DEPARTMENT OF INFORMATION TECHNOLOGY
MINI PROJECT – U23IT705 (2026-27)

DATE:

**MULTI-PILLAR DEEPPAKE DETECTION BASED ON VISION TRANSFORMER
AND ERROR LEVEL ANALYSIS**

ABSTRACT :

Generative AI has transformed the digital media threat landscape, enabling attacks that range from face-swapped videos and event imagery to AI-generated documents, voice clones, and forged financial or health records. Conventional detectors, which are predominantly single-modality and pixel-based, struggle with domain shift, compression, and novel generators, making them unreliable in real-world, cross-platform settings and leaving a critical need for a unified, trustworthy verification framework. To address these limitations, this work proposes SDD-HEVT, a hybrid spatial deepfake detection approach that combines a Vision Transformer backbone with Error Level Analysis for fine-grained tampering localization, augmented by three deterministic forensic pillars: remote photoplethysmography for physiological liveness, Benford's Law-based statistical auditing for document and image integrity, and 3D facial-geometry symmetry checks for structural consistency. These heterogeneous signals are integrated through an explainable, weighted fusion engine designed so that physically grounded evidence can confidently override uncertain learned predictions when faced with out-of-distribution or previously unseen generation techniques. The resulting ideology is a multi-domain, mobile-to-cloud verification pipeline that prioritizes robustness, interpretability, and cross-generator generalization over dataset-specific accuracy, aiming to establish a principled foundation for next-generation synthetic media authentication.

PROJECT COORDINATOR